Comparasion and Reflection of Pricing Method

P/E Ratio

- P/E=price/earning per share (EPS)
- Price
  - normally it is current price
  - someone also use the average price of the past 6 months
- EPS
  - time varies
    - » recent
    - » past four seasons
    - » the coming one year

Understand P/E Ratio

- Why do P/E ratios differ from nations, industries, companies?

  averagely, china > USA.

  high-tech company usually have higher P/E

- What factors decide P/E ratio?

- How to know whether the current P/E ratio sustains?

Back to DCF Model——Gordon Growth Model

- Gordon model, also known as constant growth model, which assumes dividends grow at a constant rate g：
  - （r 〉 g）

$$P_0 = \frac{D_0(1+g)}{(1+r)} + \frac{D_0(1+g)^2}{(1+r)^2} + \ldots = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

$$\frac{P_0}{E_0} = \frac{\frac{D_0}{E_0}(1+g)}{r-g} = \frac{z(1+g)}{r-g}$$

related to growth rate.

Life Cycle and Growth Rate at Different Stages

TABLE 18.2

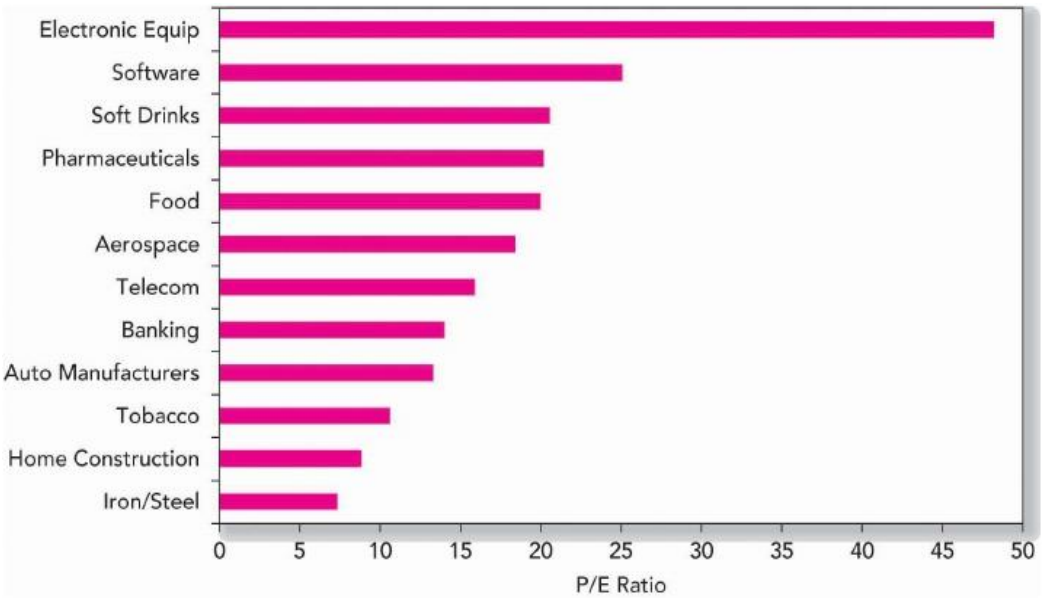
Financial ratios in two industries

	Return on Assets	Payout Ratio	Growth Rate 2005–2008
Computer Software			
Adobe Systems	21.5%	1.0%	8.2%
Cognizant	19.0	0.0	22.8
Compuware	10.5	0.0	17.6
Intuit	19.0	0.0	8.0
Microsoft	31.5	35.0	15.4
Novell	8.5	0.0	51.8
Oracle	33.0	0.0	18.6
Red Hat	17.0	0.0	17.6
Parametric Tech	20.0	0.0	33.9
SAP	22.5	18.0	13.8
Median	19.5%	0.0 → reinvest	17.6% ⇒ growth rate ↑
Electric Utilities			
Central Hudson G&E	6.0%	78.0%	5.1%
Central Vermont	7.5	60.0	8.0
Consolidated Edison	5.0	75.0	1.0
Duquesne Light	8.0	85.0	7.7
Energy East	6.0	74.0	4.1
Northeast Utilities	5.0	59.0	14.0
Nstar	8.5	61.0	3.2
Pennsylvania Power	11.0	52.0	9.3
Public Services Enter.	7.0	62.0	1.7
United Illuminating	5.0	113.0	1.3
Median	6.5%	68.0%	4.6%

Source: *Value Line Investment Survey*, 2006. Reprinted with permission of Value Line Investment Survey. © 2006 Value Line Publishing, Inc. All rights reserved.

Growth Opportunity and P/E Ratio

- the more growth opportunities, the higher P/E



P/E Ratio

- From the definition, P/E ratio is affected by
  - growth rate
  - discounted rate
  - dividend payout ratio (or reinvestment ratio)

$$P_0 = \frac{D_0(1+g)}{(1+r)} + \frac{D_0(1+g)^2}{(1+r)^2} + \ldots = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

$$\frac{P_0}{E_0} = \frac{\frac{D_0}{E_0}(1+g)}{r-g} = \frac{z(1+g)}{r-g}$$

What kind of Chinese companies have high P/E ratio?

- PE=24.08-7.50*beta+2.62*payratio+6.95*exp_5y_growth_in_EPS-1.19*std

t-statistic →（7.23）（2.58）（3.43）（0.45）

(>1.96, ✓) correlated with return (risk)

R-square：20%

- Conclusion：the Chinese companies with high P/E
  - low beta
  - relatively high dividend payout ratio
  - high expected growth rate on profit

  - but have no close relationship with
    - » volatility of stocks

Analysis of Target

- Hard to find appropriate targets sometimes
- Instead of looking for appropriate targets, try to study the relationship between P/E ratio and other factors
- Regression is also a way

American Beverage Company

Company Name	Trailing PE	Expected Growth	Standard Deviation
Coca-Cola Bottling	29.18	9.50%	20.58%
Molson Inc. Ltd. 'A'	43.65	15.50%	21.88%
Anheuser-Busch	24.31	11.00%	22.92%
Corby Distilleries Ltd.	16.24	7.50%	23.66%
Chalone Wine Group Ltd.	21.76	14.00%	24.08%
Andres Wines Ltd. 'A'	8.96	3.50%	24.70%
Todhunter Int'l.	8.94	3.00%	25.74%
Brown-Forman 'B'	10.07	11.50%	29.43%
Coors (Adolph) 'B'	23.02	10.00%	29.52%
PepsiCo, Inc. 百事	33.00	10.50%	31.35%
Coca-Cola	44.33	19.00%	35.51%
Boston Beer 'A'	10.59	17.13%	39.58%
Whitman Corp.	25.19	11.50%	44.26%
Mondavi (Robert) 'A'	16.47	14.00%	45.84%
Coca-Cola Enterprises	37.14	27.00%	51.34%
Hansen Natural Corp.	9.70	17.00%	62.45%

Source: Value Line Database.

Expected growth: the expected growth rate in the next five years

Standard deviation: volatility of stocks

Is Coca-Cola Expensive?

$$PE = 20.87 - 63.98\ \text{Standard deviation} + 183.24\ \text{Expected growth} \qquad R^2 = 51\%$$
$$[3.01]\quad [2.63] \qquad\qquad [3.66]$$

$$\text{Predicted } PE_{\text{Coca-Cola}} = 20.87 - 63.98(.3551) + 183.24(.19) = 32.97$$

market : 44%

based on this model, 可口 has been 高估.

What can explain the P/E ratio of a country?

undervalued → (want to invest in)

Country	PE Ratio	Interest Rates	GDP Real Growth	Country Risk
Argentina	14	18.00%	2.50%	45
Brazil	21	14.00%	4.80%	35
Chile	25	9.50%	5.50%	15
Hong Kong	20	8.00%	6.00%	15
India	17	11.48%	4.20%	25
Indonesia	15	21.00%	4.00%	50
Malaysia	14	5.67%	3.00%	40
Mexico	19	11.50%	5.50%	30
Pakistan	14	19.00%	3.00%	45
Peru	15	18.00%	4.90%	50
Phillipines	15	17.00%	3.80%	45
Singapore	24	6.50%	5.20%	5
South Korea	21	10.00%	4.80%	25
Thailand	21	12.75%	5.50%	25
Turkey	12	25.00%	2.00%	35
Venezuela	20	15.00%	3.50%	45

What can explain the P/E ratio of a country?

- PE = 16.16 - 7.94 Interest Rates + 154.40 Growth in GDP - 0.1116 Country Risk

  R Squared = 73%

- Conclusion
  - a country would have a high P/E if
    - » it has low interest rates
    - » it has high GDP growth rate
    - » it has low risks

What can explain the P/E ratio of a country?

Country	PE Ratio	Interest Rates	GDP Real Growth	Country Risk	Predicted PE
Argentina	14	18.00%	2.50%	45	13.57
Brazil	21	14.00%	4.80%	35	18.55
Chile	25	9.50%	5.50%	15	22.22
Hong Kong	20	8.00%	6.00%	15	23.11
India	17	11.48%	4.20%	25	18.94
Indonesia	15	21.00%	4.00%	50	15.09
Malaysia	14	5.67%	3.00%	40	15.87
Mexico	19	11.50%	5.50%	30	20.39
Pakistan	14	19.00%	3.00%	45	14.26
Peru	15	18.00%	4.90%	50	16.71
Phillipines	15	17.00%	3.80%	45	15.65
Singapore	24	6.50%	5.20%	5	23.11
South Korea	21	10.00%	4.80%	25	19.98
Thailand	21	12.75%	5.50%	25	20.85
Turkey	12	25.00%	2.00%	35	13.35
Venezuela	20	15.00%	3.50%	45	15.35

PEG

- Is a company having a high P/E ratio due to its high price or high growth?

- The definition of PEG

  = P/E ratio / expected growth rate

$$\frac{\frac{D_0}{E_0}(1+g)}{1-g} \Big/ g$$

Which one is more expensive, Pepsi or Coca-Cola?

Company Name	Trailing PE	Expected Growth	Standard Deviation
Coca-Cola Bottling	29.18	9.50%	20.58%
Molson Inc. Ltd. 'A'	43.65	15.50%	21.88%
Anheuser-Busch	24.31	11.00%	22.92%
Corby Distilleries Ltd.	16.24	7.50%	23.66%
Chalone Wine Group Ltd.	21.76	14.00%	24.08%
Andres Wines Ltd. 'A'	8.96	3.50%	24.70%
Todhunter Int'l.	8.94	3.00%	25.74%
Brown-Forman 'B'	10.07	11.50%	29.43%
Coors (Adolph) 'B'	23.02	10.00%	29.52%
PepsiCo, Inc.	33.00	10.50%	31.35%
Coca-Cola → marketing	44.33	19.00%	35.51%
Boston Beer 'A'	10.59	17.13%	39.58%
Whitman Corp.	25.19	11.50%	44.26%
Mondavi (Robert) 'A'	16.47	14.00%	45.84%
Coca-Cola Enterprises	37.14	27.00%	51.34%
Hansen Natural Corp.	9.70	17.00%	62.45%

Which one is more expensive, Pepsi or Coca-Cola?

The PE ratios and expected growth rates in EPS over the next five years, based on consensus estimates from analysts, for the firms that are categorized as beverage firms are summarized in the following table:

Company Name	*Trailing PE*	*Expected Growth*	*Standard Deviation*	*PEG*
Coca-Cola Bottling	29.18	9.50%	20.58%	3.07
Molson Inc. Ltd. 'A'	43.65	15.50%	21.88%	2.82
Anheuser-Busch	24.31	11.00%	22.92%	2.21
Corby Distilleries Ltd.	16.24	7.50%	23.66%	2.16
Chalone Wine Group Ltd.	21.76	14.00%	24.08%	1.55
Andres Wines Ltd. 'A'	8.96	3.50%	24.70%	2.56
Todhunter Int'l.	8.94	3.00%	25.74%	2.98
Brown-Forman 'B'	10.07	11.50%	29.43%	0.88
Coors (Adolph) 'B'	23.02	10.00%	29.52%	2.30
PepsiCo, Inc.	33.00	10.50%	31.35%	3.14
Coca-Cola	44.33	19.00%	35.51%	2.33
Boston Beer 'A'	10.59	17.13%	39.58%	0.62
Whitman Corp.	25.19	11.50%	44.26%	2.19
Mondavi (Robert) 'A'	16.47	14.00%	45.84%	1.18
Coca-Cola Enterprises	37.14	27.00%	51.34%	1.38
Hansen Natural Corp.	9.70	17.00%	62.45%	0.57
Average	22.66	12.60%	33.30%	2.00

from PEG角度，可口更便宜

P/B Ratio

- PB=price/book value of asset per share
  - Pros
  - Cons

① when E → 0 or negative
PE ratio no sense.
② some industry E 变动大 (oil)

interest rate ↑, 对 bank 利好 / 利空?
① 需要更多中央储备金 (可放贷少)
② 贷长 (利率一下无法调). 存款利率马上↑ (贷差值↓)

Factors Affecting P/B Ratio

$$P_0 = \frac{D_1}{(k - g_n)}$$

$$\frac{P}{BV} = \frac{ROE \times z}{k - g_n}$$

ROE × BV × Z = D_1.

$$\frac{P}{BV} = \frac{ROE - g_n}{k - g_n}$$

ROE − ROE × Z = g

k = discounted rate； z = dividend payout ratio； P_0 = price；

g_n = the growth rate of earning and dividends

D_1 = dividend next year, BV=book value

Select Stocks through P/B

The following table reports on the price-to-book ratios for integrated oil companies listed in th United States in September 2000:

Company Name	*Ticker Symbol*	*Price-to-Book Ratio*	*Return on Equity*	*Standard Deviation*
Crown Central Petroleum "A"	CNPA	0.29	–14.60%	59.36%
Giant Industries	GI	0.54	7.47%	38.87%
Harken Energy Corp.	HEC	0.64	–5.83%	56.51%
Getty Petroleum Mktg.	GPM	0.95	6.26%	58.34%
Pennzoil–Quaker State	PZL	0.95	3.99%	51.06%
Ashland Inc.	ASH	1.13	10.27%	21.77%
Shell Transport	SC	1.45	13.41%	31.61%
USX–Marathon Group	MRO	1.59	13.42%	45.31%
Lakehead Pipe Line	LHP	1.72	13.28%	19.56%
Amerada Hess	AHC	1.77	16.69%	26.89%
Tosco Corp.	TOS	1.95	15.44%	34.51%
Occidental Petroleum	OXY	2.15	16.68%	39.47%
Royal Dutch Petroleum	RD	2.33	13.41%	29.81%
Murphy Oil Corp.	MUR	2.40	14.49%	27.80%
Texaco Inc.	TX	2.44	13.77%	27.78%
Phillips Petroleum	P	2.64	17.92%	29.51%
Chevron Corp.	CHV	3.03	15.69%	26.44%
Repsol-YPF ADR	REP	3.24	13.43%	26.82%
Unocal Corp.	UCL	3.53	10.67%	34.90%
Kerr-McGee Corp.	KMG	3.59	28.88%	42.47%
Exxon Mobil Corp.	XOM	4.22	11.20%	19.22%
BP Amoco ADR	BPA	4.66	14.34%	27.00%
Clayton Williams Energy	CWEI	5.57	31.02%	26.31%
Average		2.30	12.23%	

Select Stocks through P/B

$$PBV = 2.21 + 8.22\ ROE - 2.63\ \text{Standard deviation} \qquad R^2 = 52\%$$
$$[2.16]\ [2.92] \qquad [1.21]$$

Company Name	*Price-to-Book Ratio*	*Predicted PBV*	*Under- or Overvalued*
Crown Central Petroleum "A"	0.29	–0.56	NMF
Giant Industries	0.54	1.80	–69.74% ⇒attract you
Harken Energy Corp.	0.64	0.24	166.59%
Getty Petroleum Mktg.	0.95	1.19	–19.67%
Pennzoil–Quaker State	0.95	1.19	–19.93%
Ashland Inc.	1.13	2.48	–54.28%
Shell Transport	1.45	2.48	–41.56%
USX–Marathon Group	1.59	2.12	–25.11%
Lakehead Pipe Line	1.72	2.78	–38.03%
Amerada Hess	1.77	2.87	–38.33%

Select Stocks through P/B

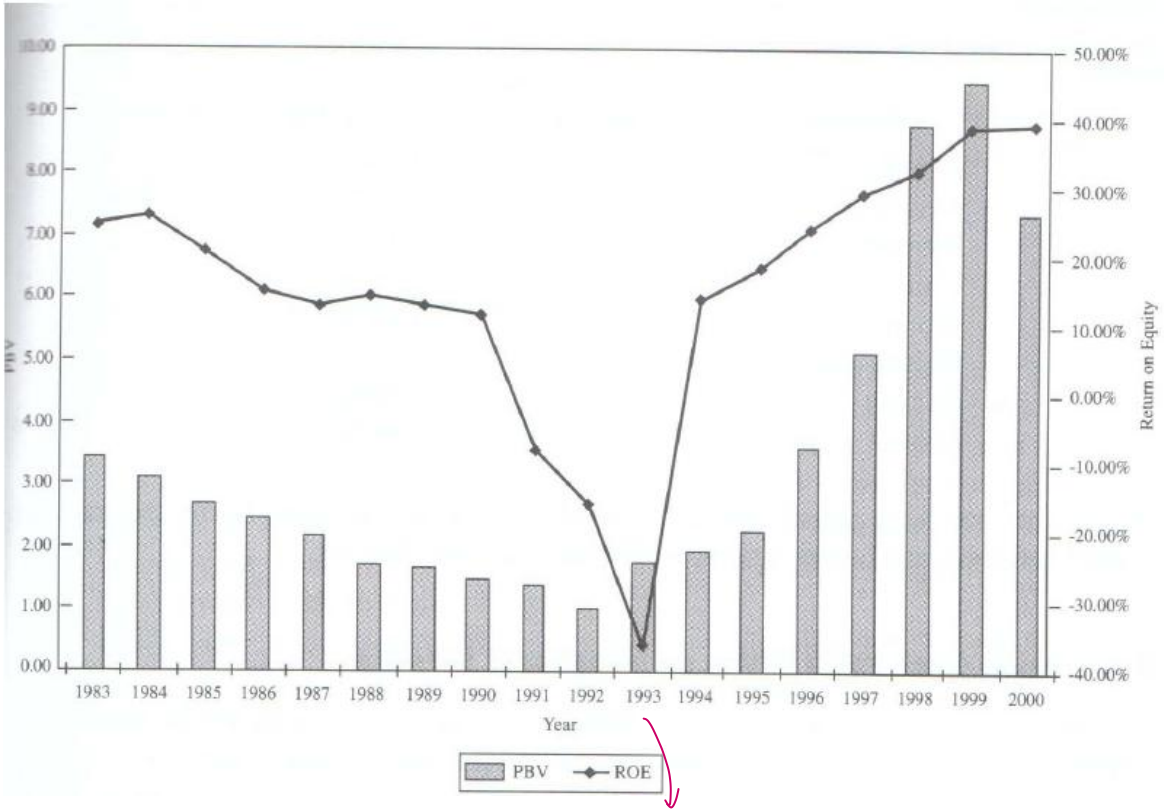
Company Name	Price-to-Book Ratio	Predicted PBV	Under- or Overvalued
Tosco Corp.	1.95	2.57	–24.09%
Occidental Petroleum	2.15	2.54	–15.27%
Royal Dutch Petroleum	2.33	2.52	–7.66%
Murphy Oil Corp.	2.40	2.67	–10.07%
Texaco Inc.	2.44	2.61	–6.47%
Phillips Petroleum	2.64	2.90	–9.17%
Chevron Corp.	3.03	2.80	8.20%
Repsol-YPF ADR	3.24	2.60	24.53%
Unocal Corp.	3.53	2.17	63.05%
Kerr-McGee Corp.	3.59	3.46	3.70%
Exxon Mobil Corp.	4.22	2.62	60.99%
BP Amoco ADR	4.66	2.67	74.03%
Clayton Williams Energy	5.57	4.06	36.92%

Which one is the most expensive?

Which one is the cheapest?

strong correlation

The P/B and Return of IBM



换了CEO (strategy)

Price/Sales Ratio

- Pros
  - less likely to be manipulated, compared to earnings and book value
  - earnings may be negative or close to zero
  - more stable than P/E
  - more suitable to price mature, cyclical company with close zero profit

周期

Price/Sales Ratio

- Cons
  - large sales is not equal to high profit
  - stock price should not tightly related to sales

- If there are two stocks having the same P/E, which one would have higher price/sales ratio?

Price/Sales Model： 1-stage Model

profit margin = E/S

$$P_0 = \frac{D_1}{(k - g_n)} \Rightarrow \frac{P}{S} = \frac{PM \times z \times (1 + g_n)}{k - g_n}$$

$D_1 = E \cdot z(1+g_n)$

$\frac{P}{S} = \frac{D_1}{S}$

k = discounted rate； z = dividend payout ratio； P_0 = price；

g_n = the growth rate of earning and dividends

D_1 = dividend next year： S= sales； PM = net profit margin=E/S

2-Stage Model

$$\frac{P_0}{S_0} = PM \times \left\{ \frac{z_n \times (1+g) \times \left[1 - \frac{(1+g)^n}{(1+k_{hg})^n} \right]}{(k_{hg} - g)} + \frac{z \times (1+g)^n \times (1+g_n)}{(k_{st} - g_n) \times (1+k_{hg})^n} \right\}$$

PM = net profit margin

g = growth rate in the previous n years

k_{hg} = discounted rate during high growth period

k_{st} = discounted rate during stable growth period

z = dividend payout ratio during high growth period

g_n = growth rate after the first n years

z_n = dividend payout after the first n years

Specific Ratio in Different Industries

Commodity Company（oil refining, gold mining companies）

$$\text{unit commodity value} = \frac{\text{Stock value + debt value - cash}}{\text{commodity inventory}}$$

If manufacturer produce products with identical quality:

$$\text{unit product value} = \frac{\text{stock value + debt value - cash}}{\text{product units（or capacity）}}$$

Specific Ratio in Different Industries

Internet company、information provider、power cable company, etc：

usually do NOT have positive earning.

$$\text{Unit user value} = \frac{\text{stock value + debt value - cash}}{\text{number of users}}$$

(PE should use big E)

Amazon：

$$\text{unit customer value} = \frac{\text{stock value + debt value- cash}}{\text{number of effective customers}}$$ ⇒ true important.

$$\text{unit visitor value} = \frac{\text{stock value + debt value - cash}}{\text{number of visitors}}$$

Company with patents: the price when Google bought Motorola

Value of Internet Flow

Company Name	*Firm Value*	*Visitors*	*Value per Visitor*
Lycos, Inc.	$ 5,396.00	5,858	$0.92
MapQuest.com Inc.	$ 604.80	6,621	$0.09
iVillage Inc.	$ 250.40	7,346	$0.03
CNET Networks	$ 1,984.30	10,850	$0.18
Ask Jeeves Inc.	$ 643.50	11,765	$0.05
Go2Net Inc.	$ 1,468.60	12,527	$0.12
LookSmart, Ltd.	$ 1,795.30	13,374	$0.13
About.com Inc.	$ 541.90	18,282	$0.03
Excite@Home	$ 7,008.20	27,115	$0.26
Yahoo! Inc.	$65,633.40	49,045	$1.34

Source: Media Metrix.

Examples of Valuation Application

苏宁

Price (RMB)	10.95
Shares	6,996,211,900
RMB'000	
Equity	76,608,520
Debt	35,348
Cash	21,568,655
EV enterprise	55,075,213
Store	962
Area (Sqm)	4,058,300
RMB	
EV/Store	57,250,741
Sales/Store	61,955,525
EV/Sqm	13,571
Sales/Sqm	14,602

国美

Price (RMB)	2.00
Shares	15,055,332,000
RMB'000	
Equity	30,110,664
Debt	3,524,909
Cash	6,029,059
EV	27,606,514
Store	726
Area (Sqm)	2,675,000
RMB	
EV/Store	38,025,501
Sales/Store	58,771,350
EV/Sqm	10,320
Sales/Sqm	15,951

→ store is different. Do not control size.

→ 1 sqm in center & country is different

÷ ↓

= EV/Sales fair comparison

Which one had higher estimated value in May 2010?

Examples of Valuation Application

沃尔玛

Price (USD)	50.02
Shares(Million)	3,786
USD Million	
Equity	189,376
Debt	33,754
Cash	7,907
EV	215,223
Store	8,416
Area (Sqm)	
RMB	
EV/Store	174,689,987
Sales/Store	328,677,519

Walmare has more large store.

Best Buy

Price (USD)	40.56
Shares(Million)	419
USD Million	
Equity	16,995
Debt	1,802
Cash	1,826
EV	16,971
Store	4,027
Area (Sqm)	
RMB	
EV/Store	28,787,385
Sales/Store	84,283,591

物美

Price (RMB)	16.37
Shares	1,251,274,116
RMB'000	
Equity	20,483,357
Debt	456,086
Cash	1,171,575
EV	19,767,868
Store	472
Area (Sqm)	542,241
RMB	
EV/Store	41,881,077
Sales/Store	24,961,883

More expensive for $\frac{EV}{Sales}$. why?

Which one is the most expensive?

① higher profit for 1 sale

② still expanding. (high growth r)

Growth Opportunity and Return

- The growth rate of American computer software companies is 17.6%, while that of electricity companies is only 4.6%. Hence should we buy stocks of computer software companies?

Buy high / low growth company ?

logically. people like high growth rate.

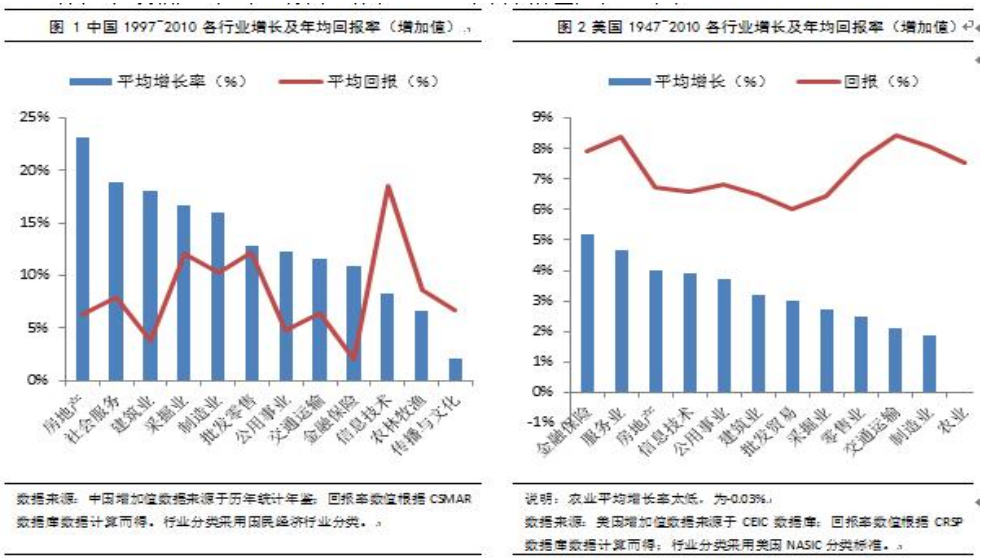
But high growth company is expensive.

(No garantee. might hurt investor heavily)

Growth Opportunity and Return

There is no obvious pattern for the growth and return rates in industries of China. The industry with high growth does not mean high return. Take agricultural and real estate industries as examples, the average annual growth rate of real estate is 23.13%, and that of agriculture is 6.69%. If you invest in real estate and agriculture in 1997, you should have returns of 6.27% and 8.58% separately in 2010. The similar things also appear in IT, retail, mining and construction industries. The returns of low growth industry are higher than that of high growth industry.

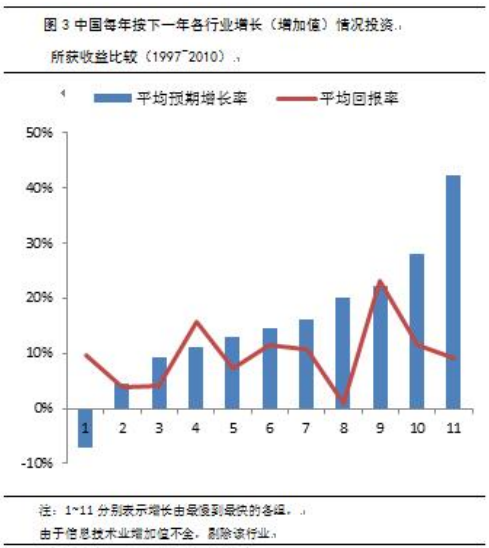
we don't see relativity btw growth & return ⇒

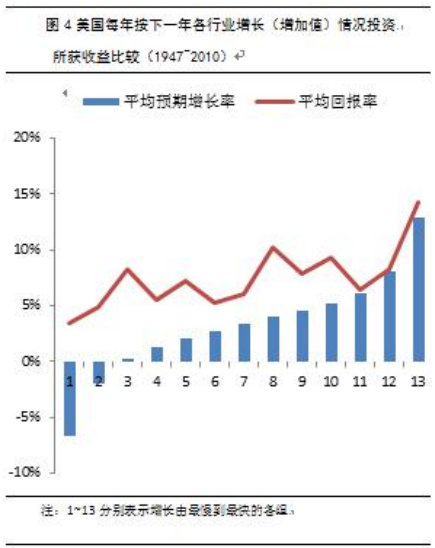


Growth Opportunity and Return

Buy the industry with high growth next year/ last year

In reality, investors may frequently buy and sell. Assume that investor can estimate the future. They buy at the beginning of each year and sell at the end of year according to the industry growth. Under the assumption of “high growth high return”, stocks in the high growth industries should have higher return.





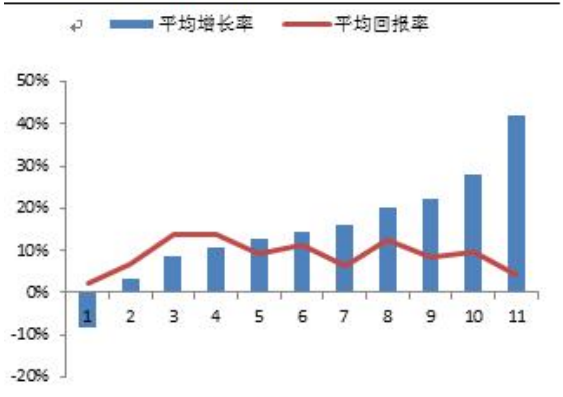
There is no obvious pattern in China, but in the US, it appears that in the industries with the highest and lowest growth rate, there is the pattern.

Jinfan Zhang

Growth Opportunity and Return

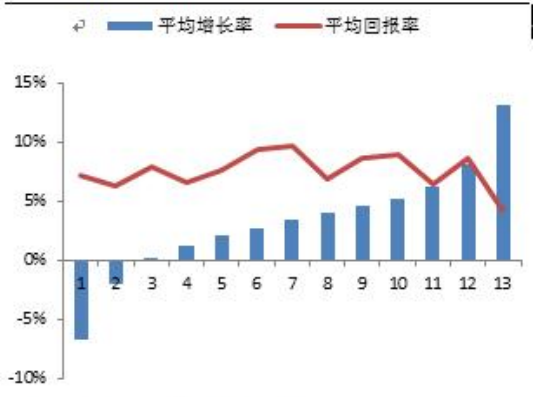
If we take a feasible strategy, which rebalances the portfolio according to the growth rate of the industry last year, no return pattern could be discovered.

图 5 中国每年按上一年各行业增长（增加值）情况投资所获收益比较（1997~2010）



注：1~11 分别表示增长由最慢到最快的各组。
由于信息技术业增加值不全，剔除该行业。

图 6 美国每年按上衣年各行业增长（增加值）情况投资所获收益比较（1947~2010）



注：1~13 分别表示增长由最慢到最快的各组。

Growth Opportunity and Return

We do not see the solid relationship between high growth and high return. Why?

Market may over expect the growth. The valuation is too high.

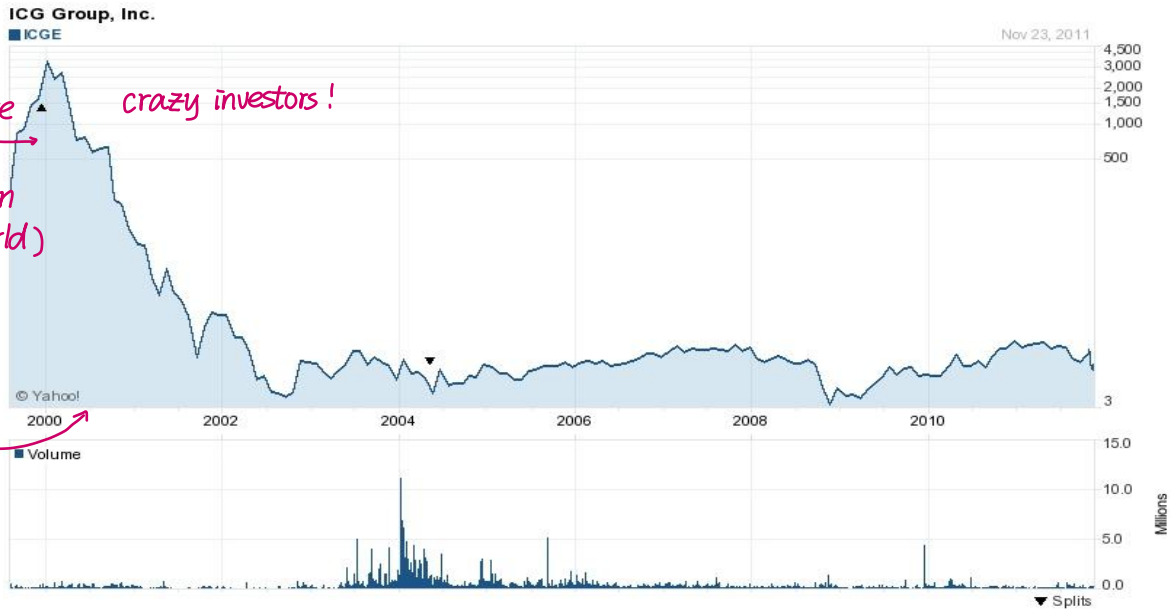
Can the stock price be estimated?

The pricing of Internet Capital Group

very high price (people believe this company can change the world)

crazy investors!

internet bubble



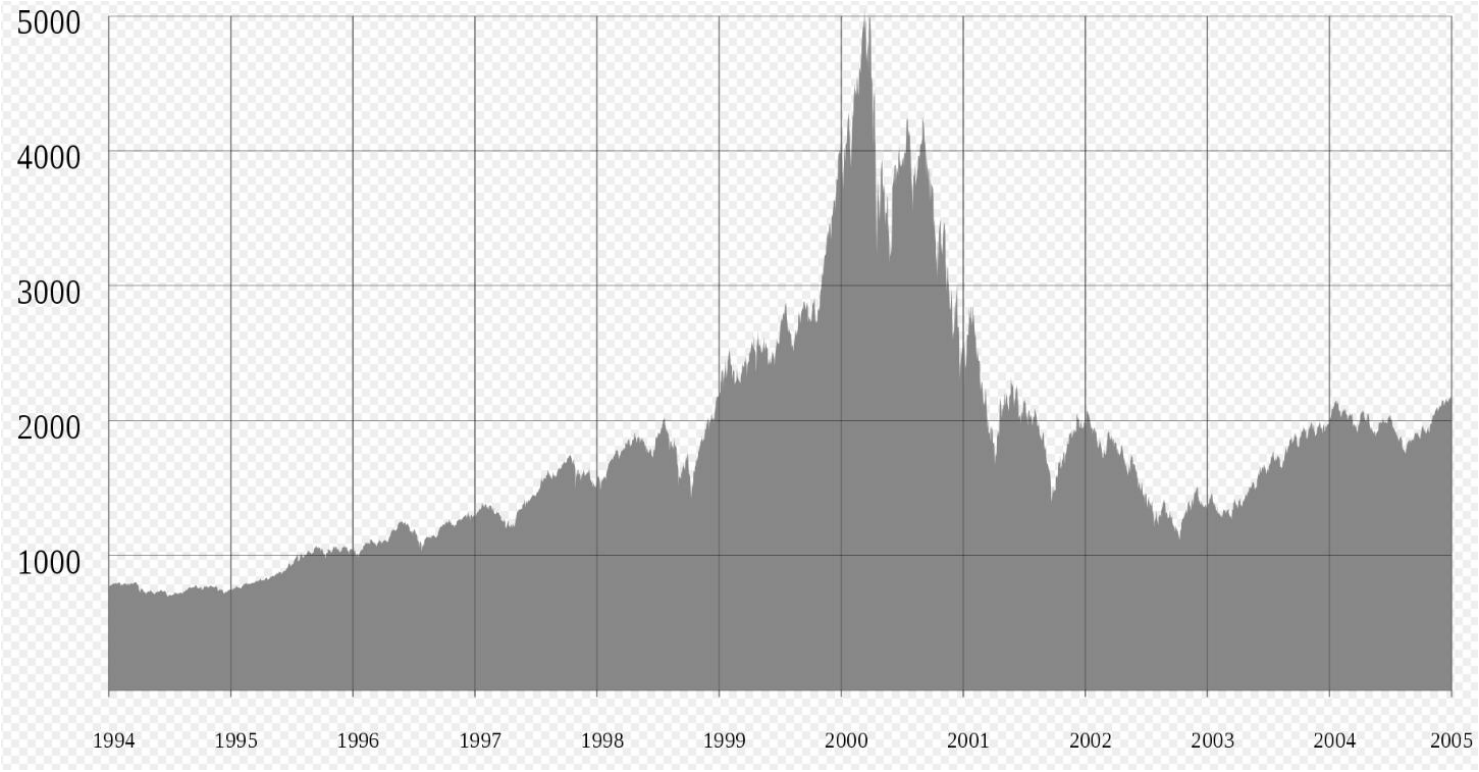
Nov 23, 2011, $7.52

张劲帆

Looking for Another Fool

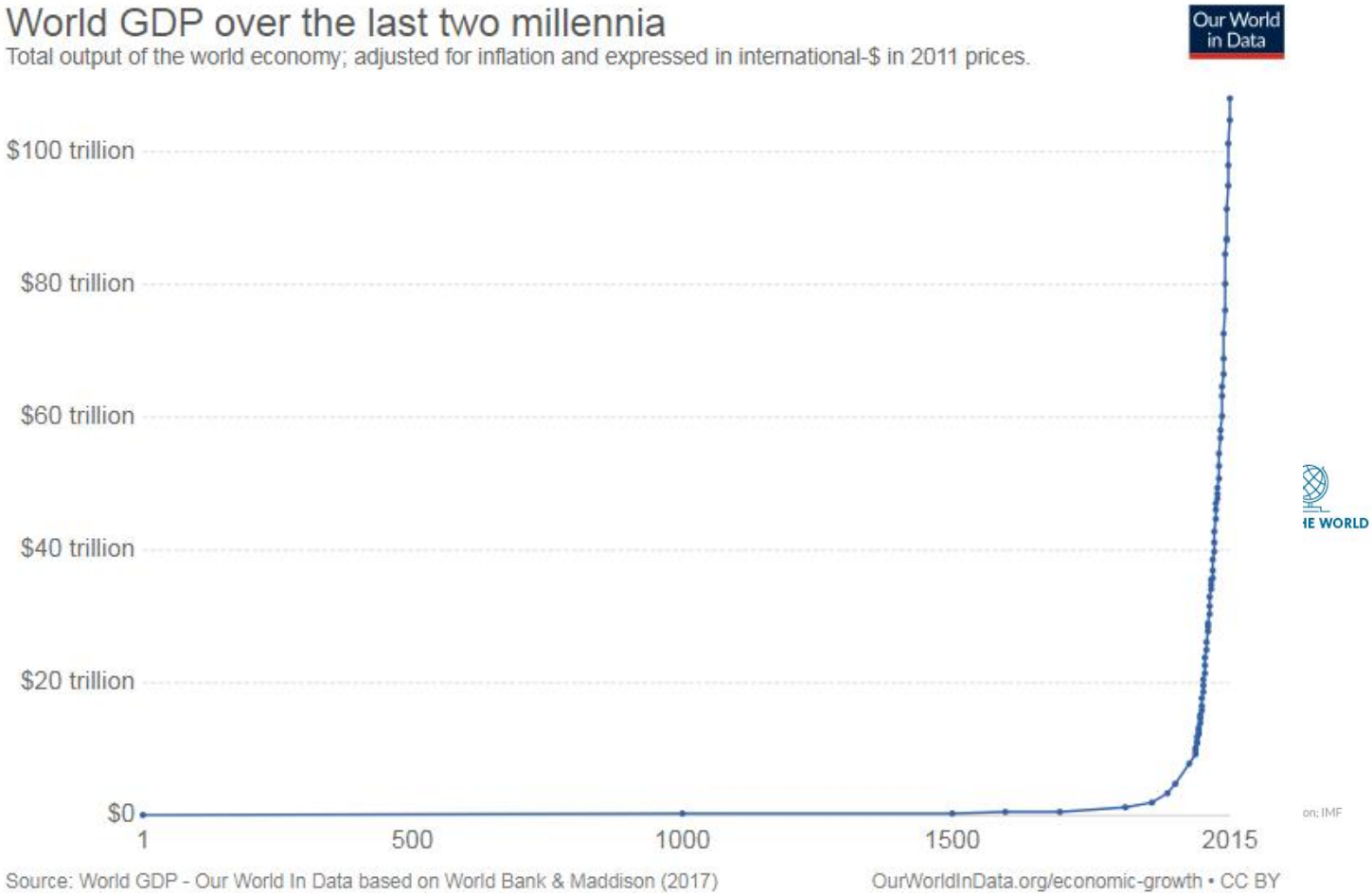
- the pricing of Internet Capital Group
  - "Pricing method is often ineffective when applied to high growth company (Henry Blodget，analyst of Merill Lynch stock said it to Internet Capital Group in Jan 2000.）
- many investors think that the stock price depend on "supply and demand"
  - supply and demand depend on expectation
  - expectation depends on the "next fool"
- in 1999
  - S&P500 index increased 18%
  - Buffet' Berkshire Hathaway dropped 23%
  - 《Time》： "Warren，what's wrong？"

The NASDAQ index spiked in the late 1990s and then fell sharply as a result of the dot-com bubble.





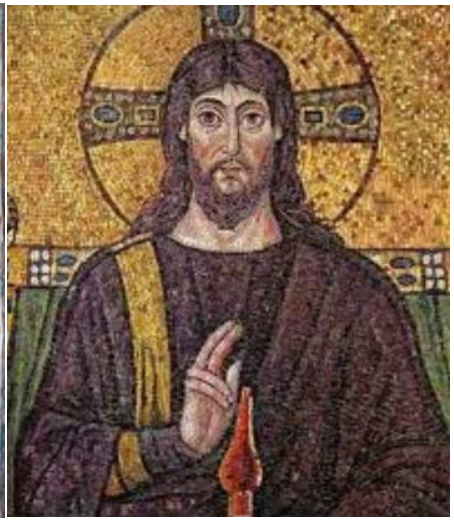
Past 20 years: the peak-to-peak compound return: 1.05^20=2.6=13000/5000
Past 40 years: from 1981-2021 compound return: 1.11^40=65=13000/200


World GDP over the last two millennia
Total output of the world economy; adjusted for inflation and expressed in international-$ in 2011 prices.
Our World in Data
$100 trillion
$80 trillion
$60 trillion
$40 trillion
$20 trillion
$0
1
500
1000
1500
2015
Source: World GDP - Our World In Data based on World Bank & Maddison (2017)
OurWorldInData.org/economic-growth • CC BY

Medieval vs Renaissance

humanism

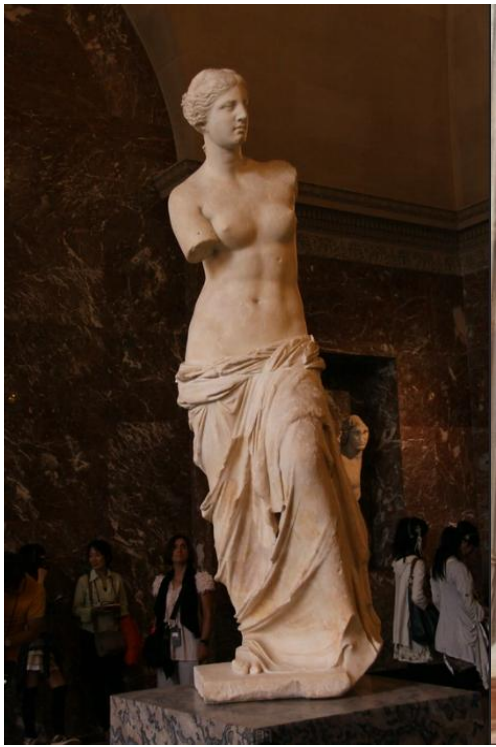


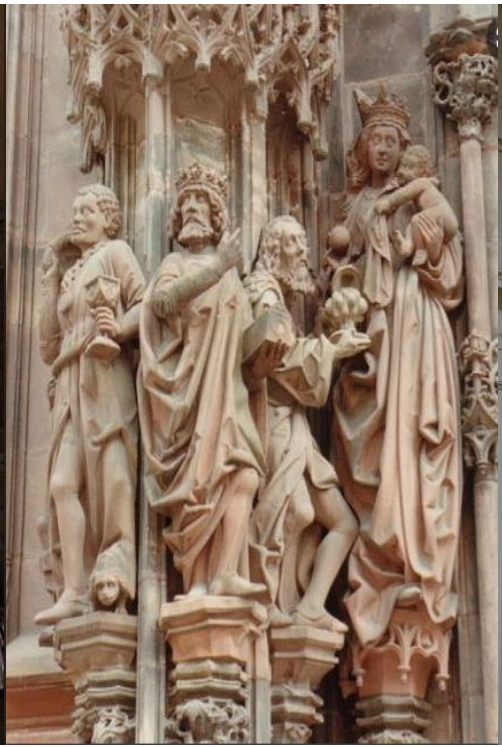






Medieval vs Renaissance







Medieval vs Renaissance

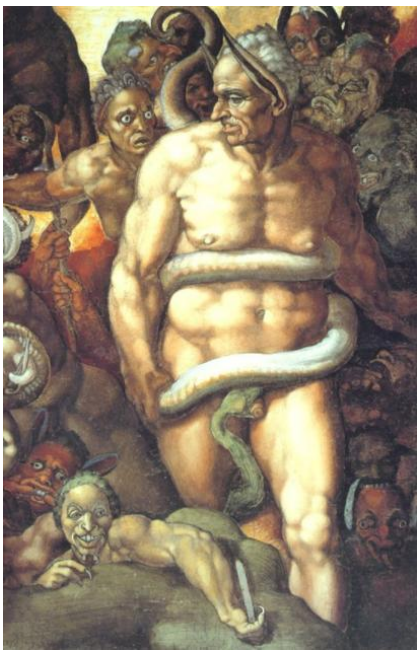
- Sistine Chapel--Michelangelo



Medieval vs Renaissance

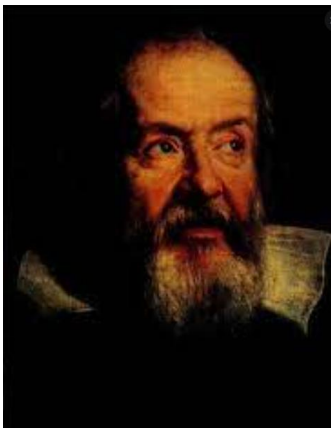


Sistine Chapel--Michelangelo

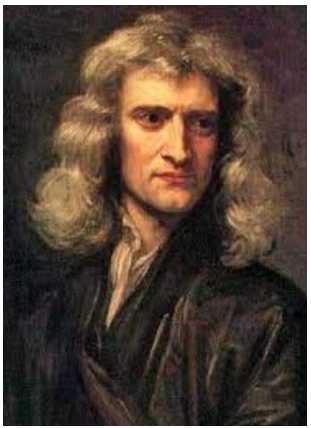


Science

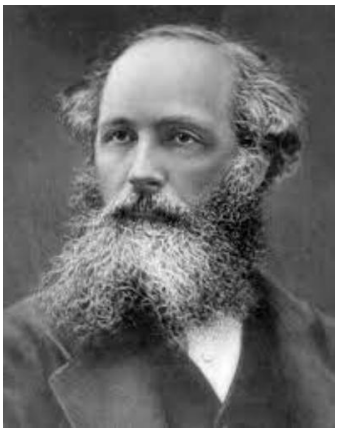
Rationalism



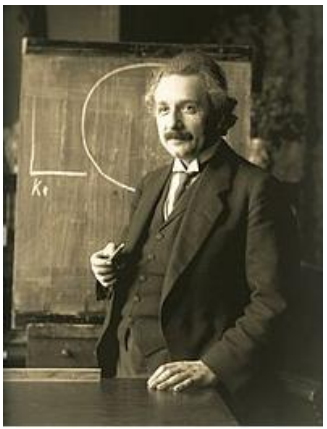
伽利略
(1564－1642)



牛顿
(1642－1727)

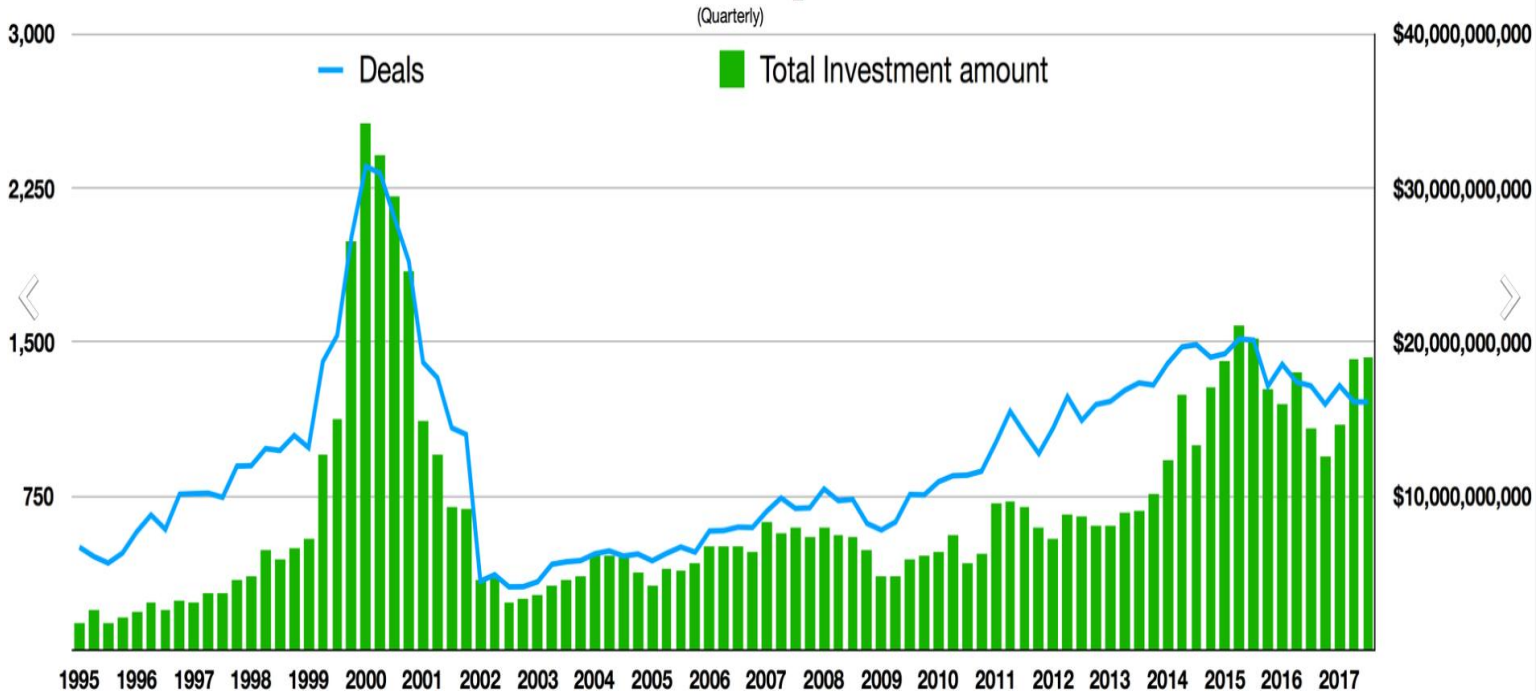


麦克斯韦
(1831－1879）



爱因斯坦
(1879－1955)

Total U.S. Venture Capital Investments



Worries of Li Guoqing



- Market value of Dangdang increased 87% on its first-day closing price, on Dec. 9, 2010, reaching $29.91/share
- The price of Dangdang on Aug 28, 2015 was $6.15/share

迷失的唯怡：哈哈哈哈哈，此女必火，李国庆你简直就不是人，三个月前好像是你嘴上抹了蜜把一本负盈利的烂帐颤微微地四处磕头求你老婆动用她以前的社会积累和关系拉你一把，做了十年的破网就快倒闭了不想死得这么难看，你别忘了这可是有很多目击证人的。

@当当网李国庆V：摇滚歌词，虚构：为做俺们生意，你们Y给出估值10-60亿，一到香港写招股书，总看韩朝开火，只写7.8亿，别TMD演戏。我大发了脾气。老婆享受辉煌路演，忘了你们为啥窃窃私喜。王八蛋们明知次日开盘就会20亿；还定价16，也就11亿。次日开盘，CFO被股价吓的尿急，我说忍了这口气，过了静默期我再操你妈了逼　原文转发(3952) | 原文评论(1815)

张劲帆

沪深300 (large size)

中证500（399905.SZ）500 middle size



Can the return of China stock market be estimated?

Time Series Estimation vs Cross Sectional Estimation

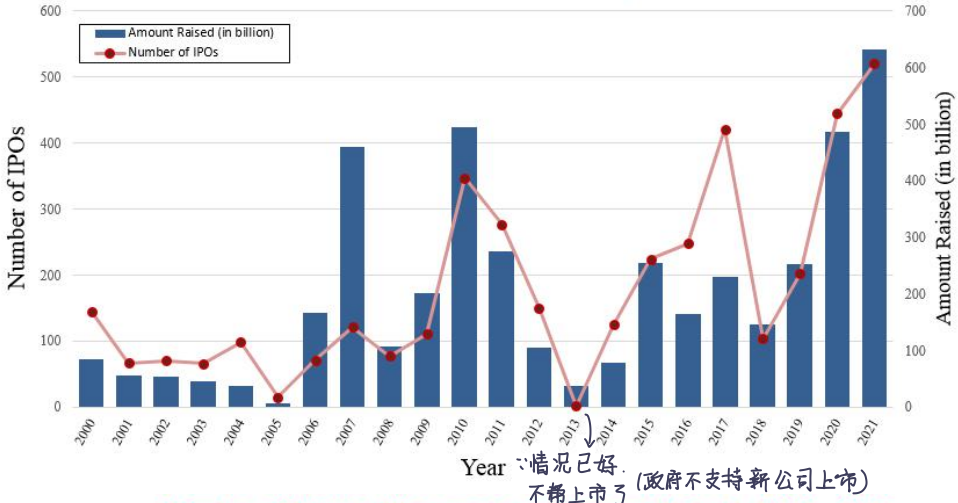
in US, many company voluntary exit stock market. (∵ costly for being public)

in China, few company want to quit. (And 上市流程复杂&时间长)

借壳上市: buy bad company, issue shares (investor actually buy the company behind)

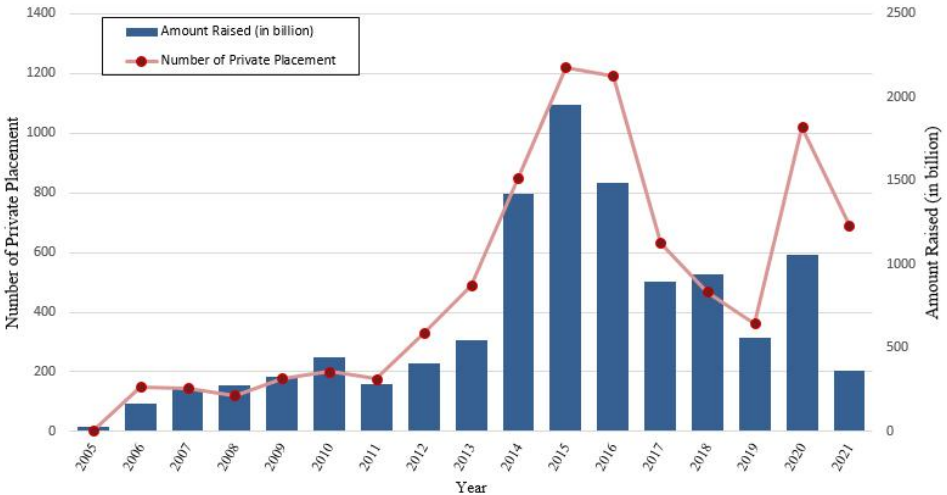


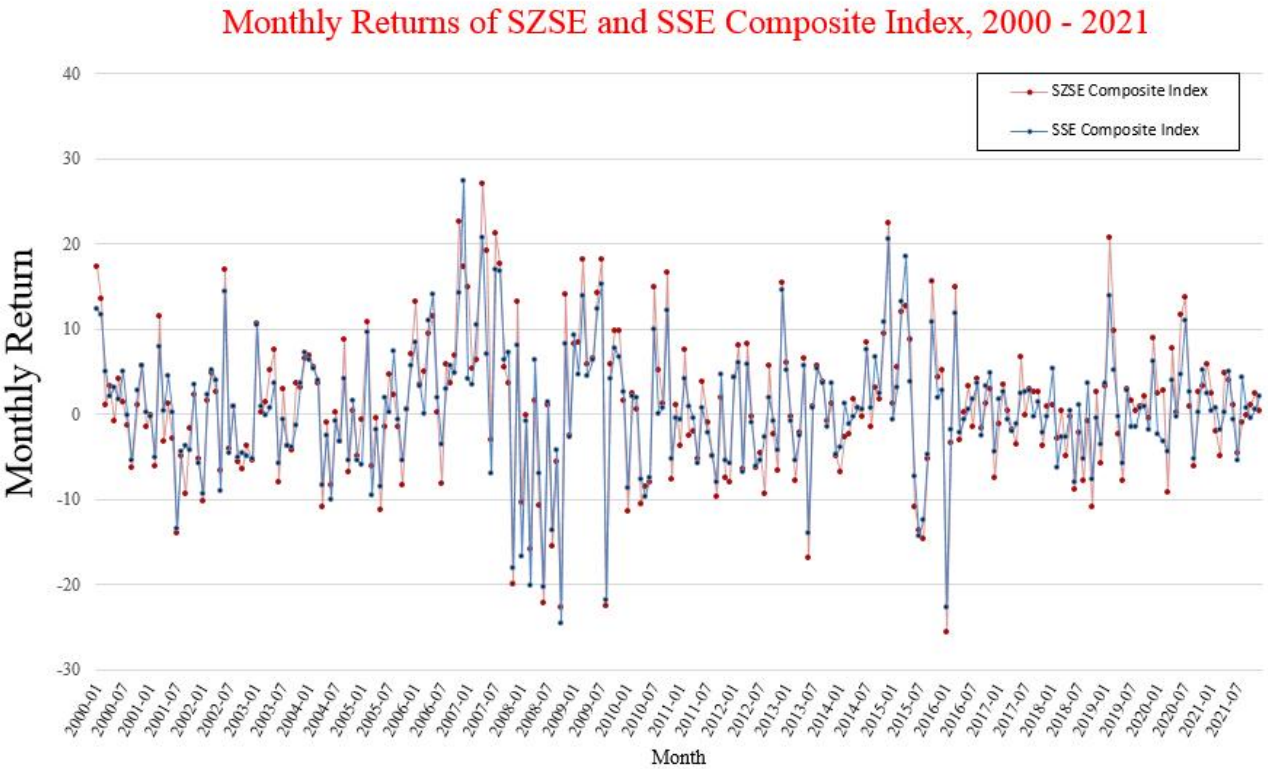
Number of IPOs and Amount Raised, 2000 - 2021



情況已好.
不都上市了 (政府不支持新公司上市)

Number of Private Placement and Amount Raised, 2005 - 2021





large → shanghai ; small size → shenzhen

SSE: mean=0.64% std=7.28%
SZSE: mean=0.90%, std=8.27%

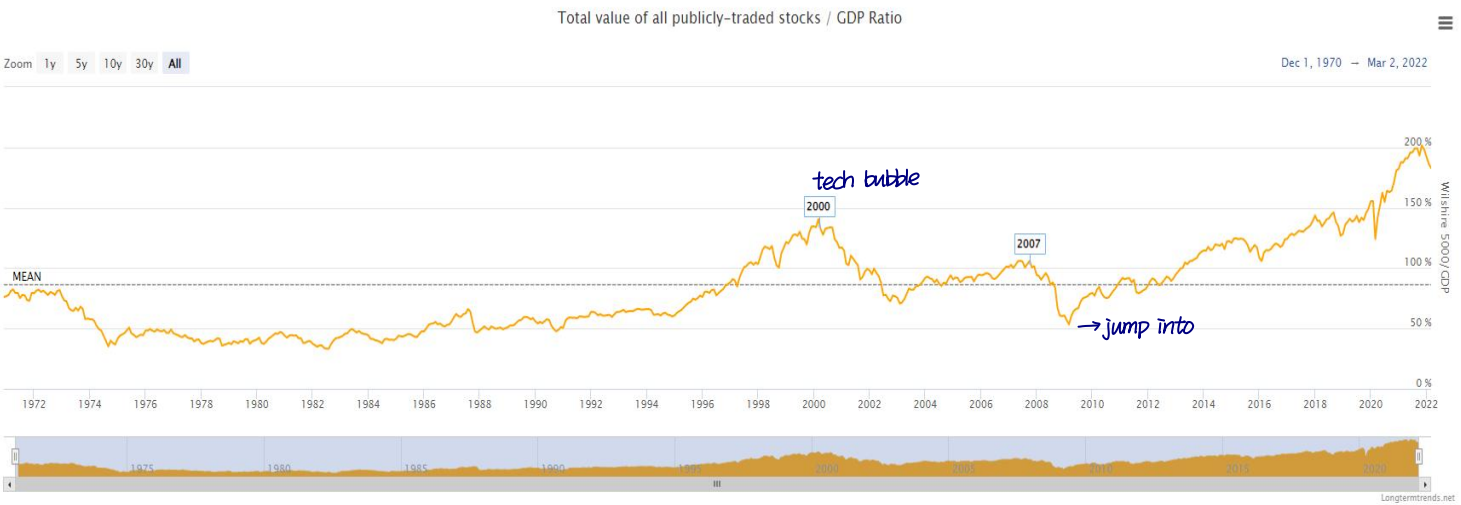
sz generate better. why?


Cumulative Returns for SZSE and SSE Composite Index, 2000 - 2021
SZSE Composite Index
SSE Composite Index
Cumulative Return (in %)
Month
450
400
350
300
250
200
150
100
50
0
-50
-100
2000-01
2000-07
2001-01
2001-07
2002-01
2002-07
2003-01
2003-07
2004-01
2004-07
2005-01
2005-07
2006-01
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2016-01
2016-07
2017-01
2017-07
2018-01
2018-07
2019-01
2019-07
2020-01
2020-07
2021-01
2021-07


Monthly PE Ratio in Shanghai Stock Exchange and Shenzhen Stock Exchange, 2007 - 2021
Shenzhen Stock Exchange
Shanghai Stock Exchange
PE Ratio (in %)
Month
0
10
20
30
40
50
60
70
80
2007-01
2007-05
2007-09
2008-01
2008-05
2008-09
2009-01
2009-05
2009-09
2010-01
2010-05
2010-09
2011-01
2011-05
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2014-09
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2015-05
2015-09
2016-01
2016-05
2016-09
2017-01
2017-05
2017-10
2018-02
2018-06
2018-10
2019-02
2019-06
2019-10
2020-02
2020-06
2020-10
2021-02
2021-06
2021-10

US stock market benchmark



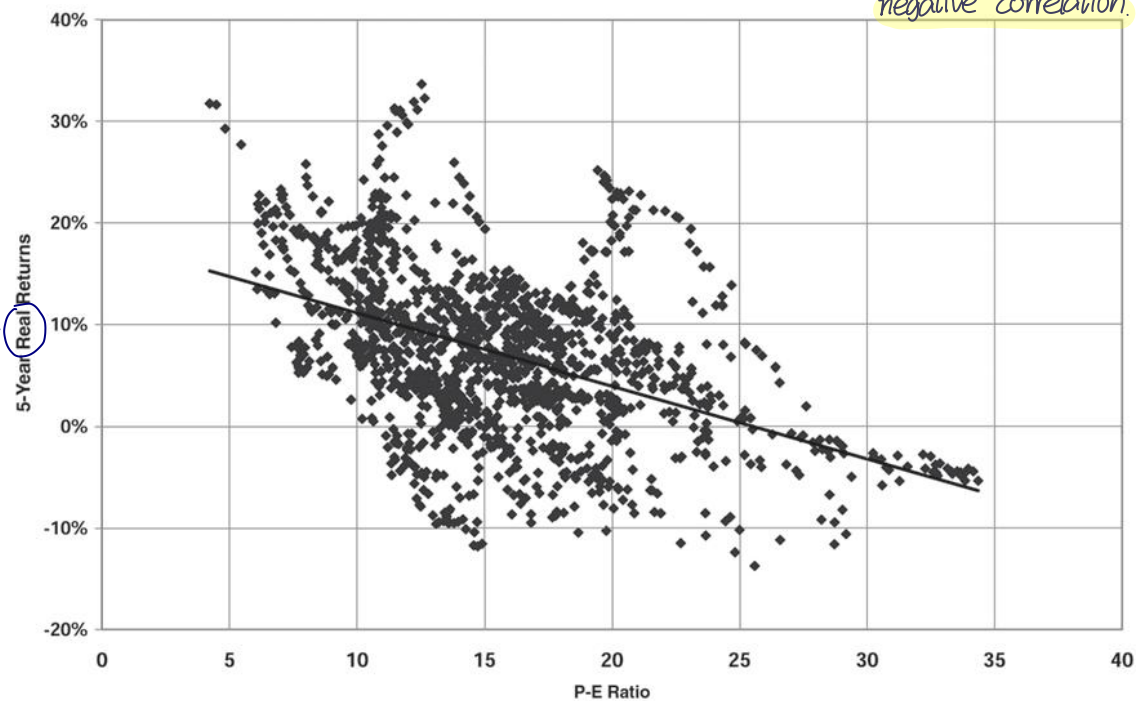


P/E Ratio and the Actual Return of Stock in the next Five Years (Siegel)

Can the return of China stock market be estimated?

current PE ↔ future return

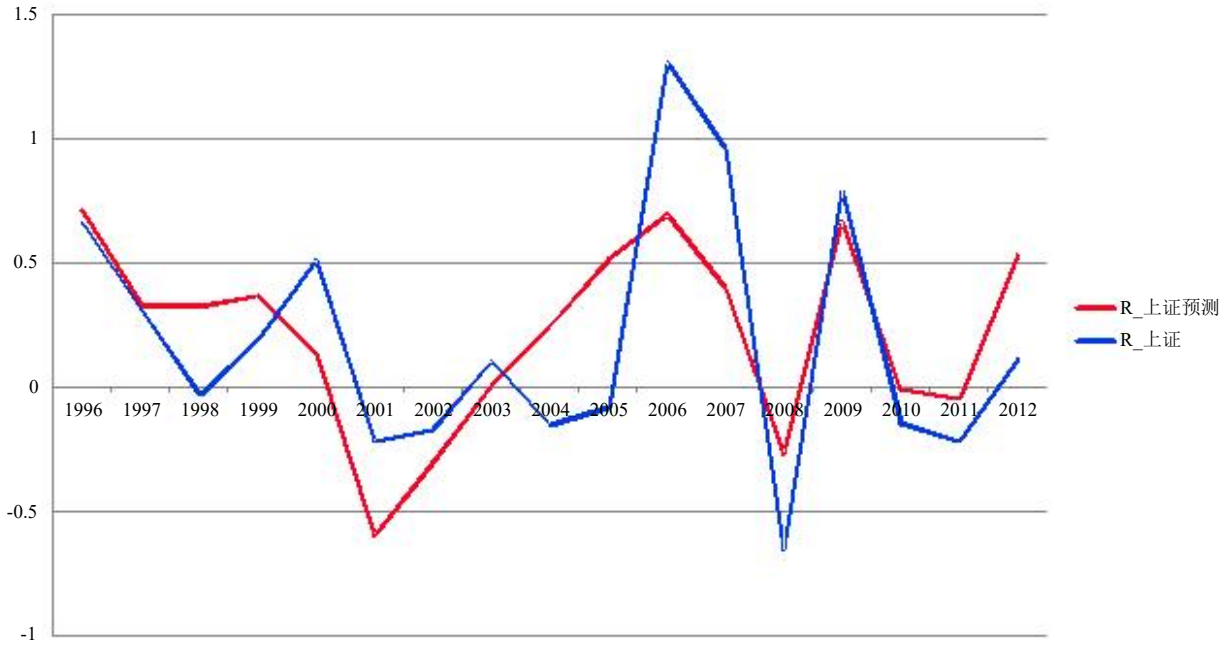
negative correlation



minus inflation

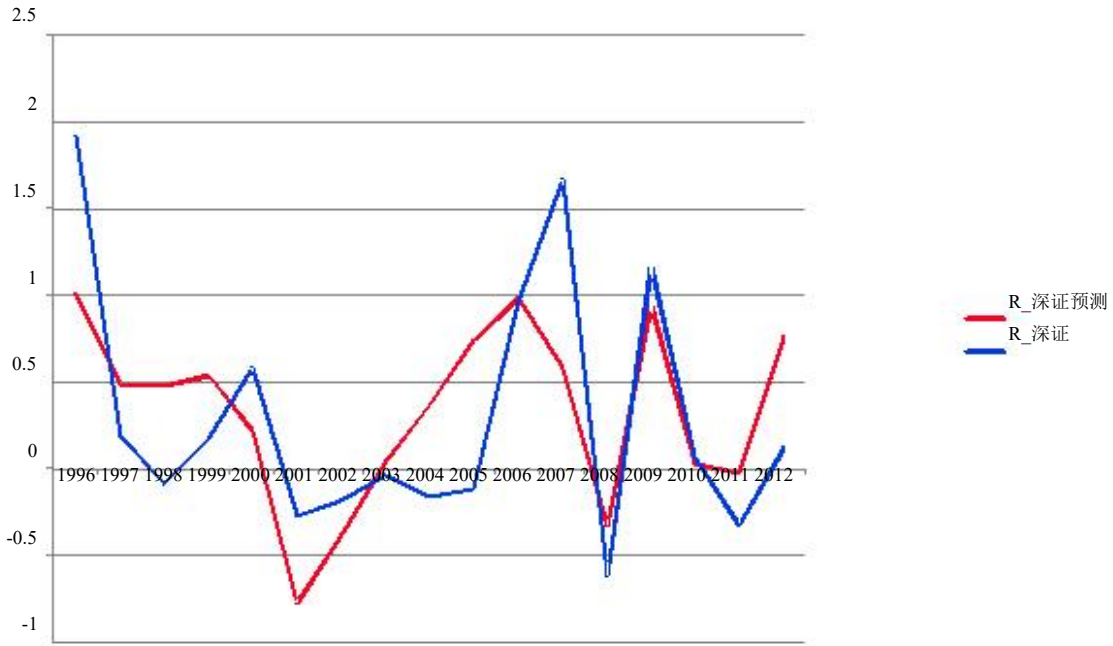
张劲帆

Chinese Market Trend: P/E ratio Estimates Return



张劲帆

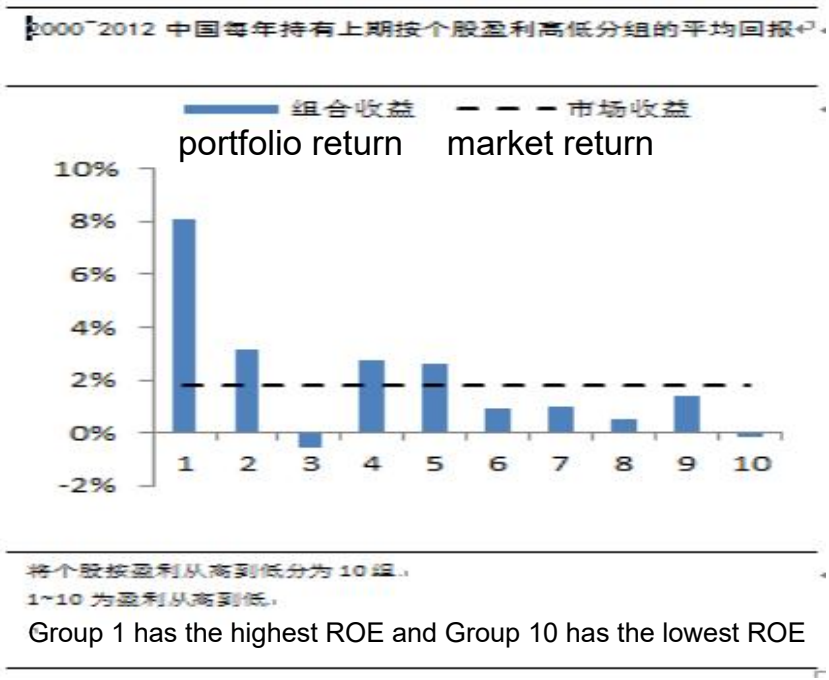
Chinese Market Trend: P/B ratio Estimates Return



张劲帆

Can the Chinese market trend be estimated?

Can the ROE of a stock estimate the future return?



Group 1 has the highest ROE and Group 10 has the lowest ROE

We divide the stocks into 10 groups (1-10) according to the ROE of the stocks at the end of each year . Then, the VW average return is calculated for each group for the following year. We repeat this process every year. Then We calculate the average return for each group.

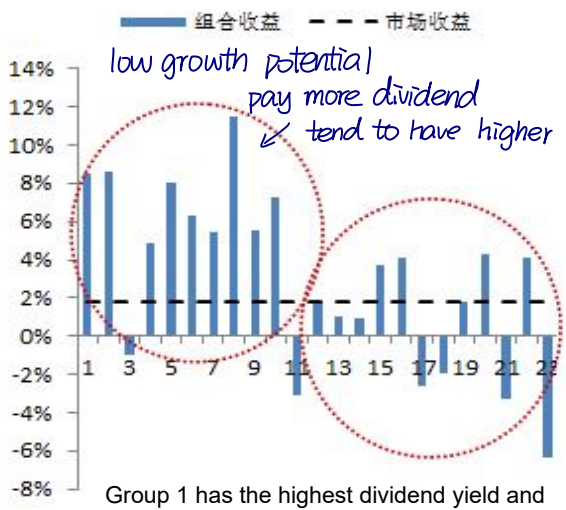
张劲帆

Dividend yield Predicts Return

Can the dividend yield predict the stock return?

At the end of each year, we rank the industries (individual stocks) with their dividend yields and divide them into 23 groups. We then calculate the average return for each group from 2000 to 2012, and compare it with the market. High dividend yield stocks beat the market.

2000~2012 中国每年持有上年按各行业股息率高低排序的平均回报。

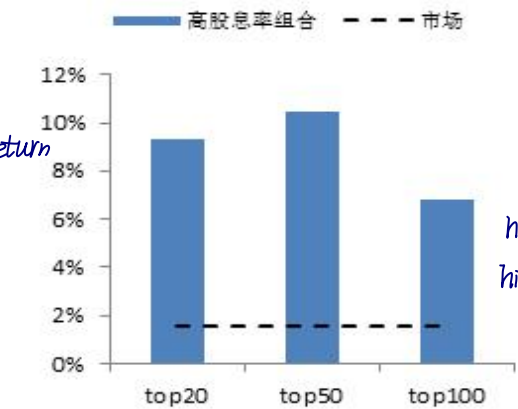


low growth potential
pay more dividend
tend to have higher return

Group 1 has the highest dividend yield and Group 10 has the lowest dividend yield

股息率为减去各行业前 5 年平均值后的值；1~23：股息率从高到低；

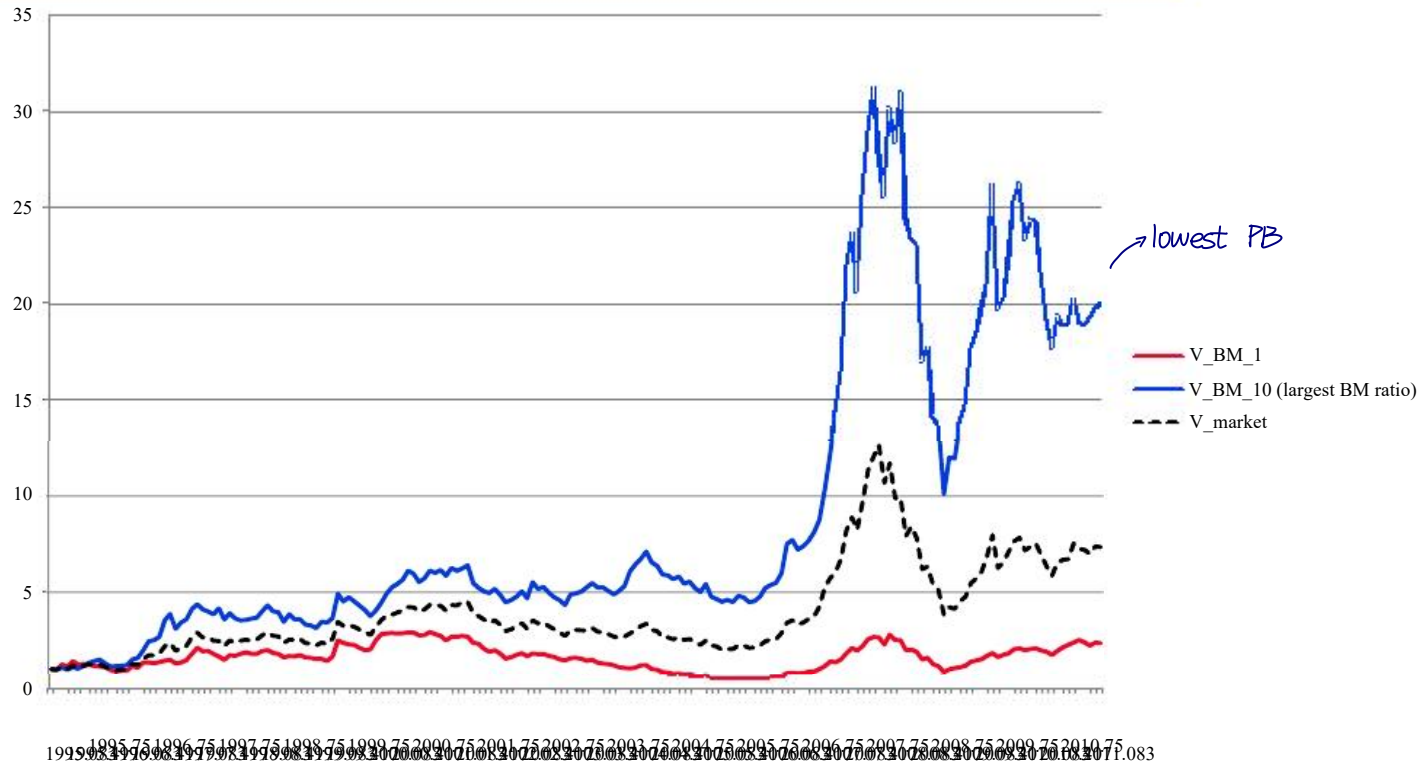
2000~2012 中国每年持有上年股息率最高的股票组合取得的回报



high dividend generate high return in China

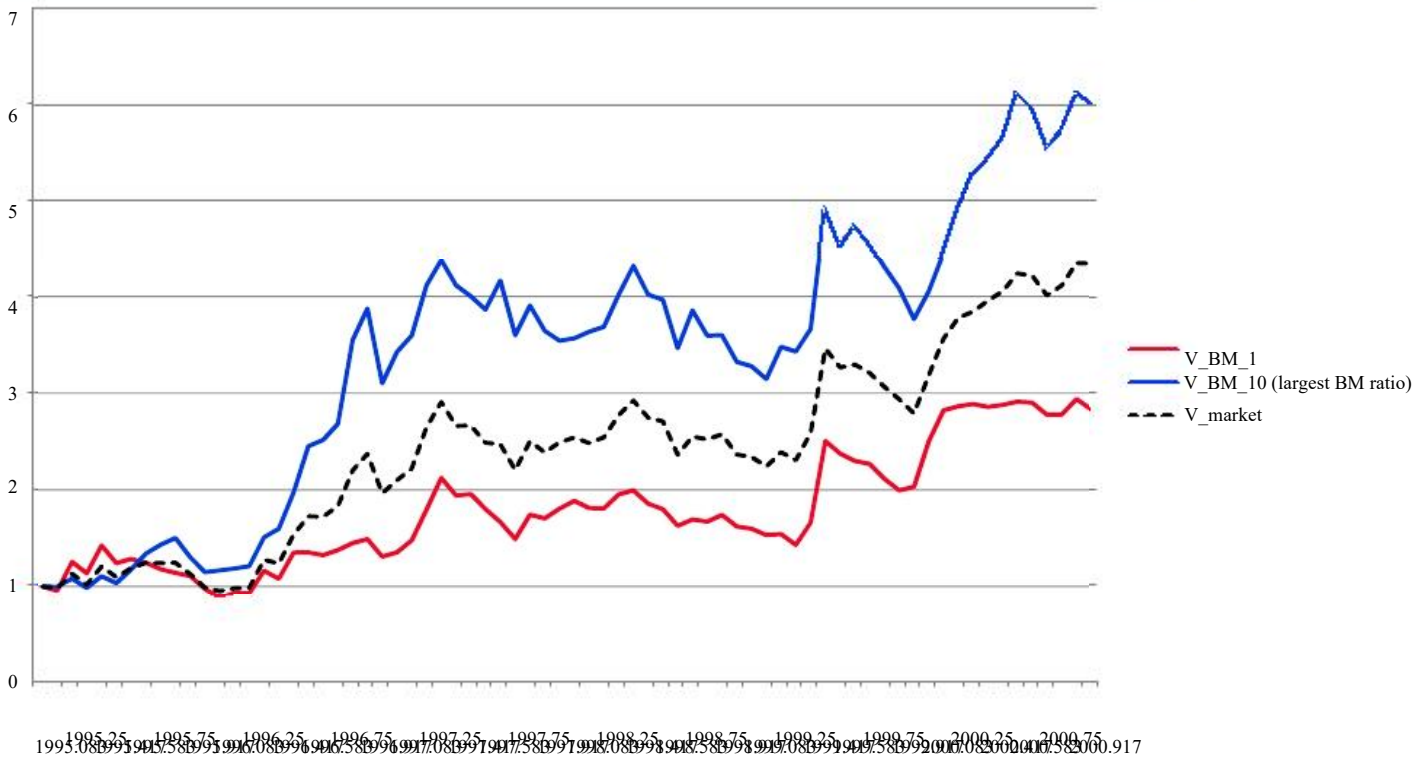
Top20 为当年股息率最高的 20 只股票，其余以此类推。

Value Stocks Won the Market: China (1995-2010)



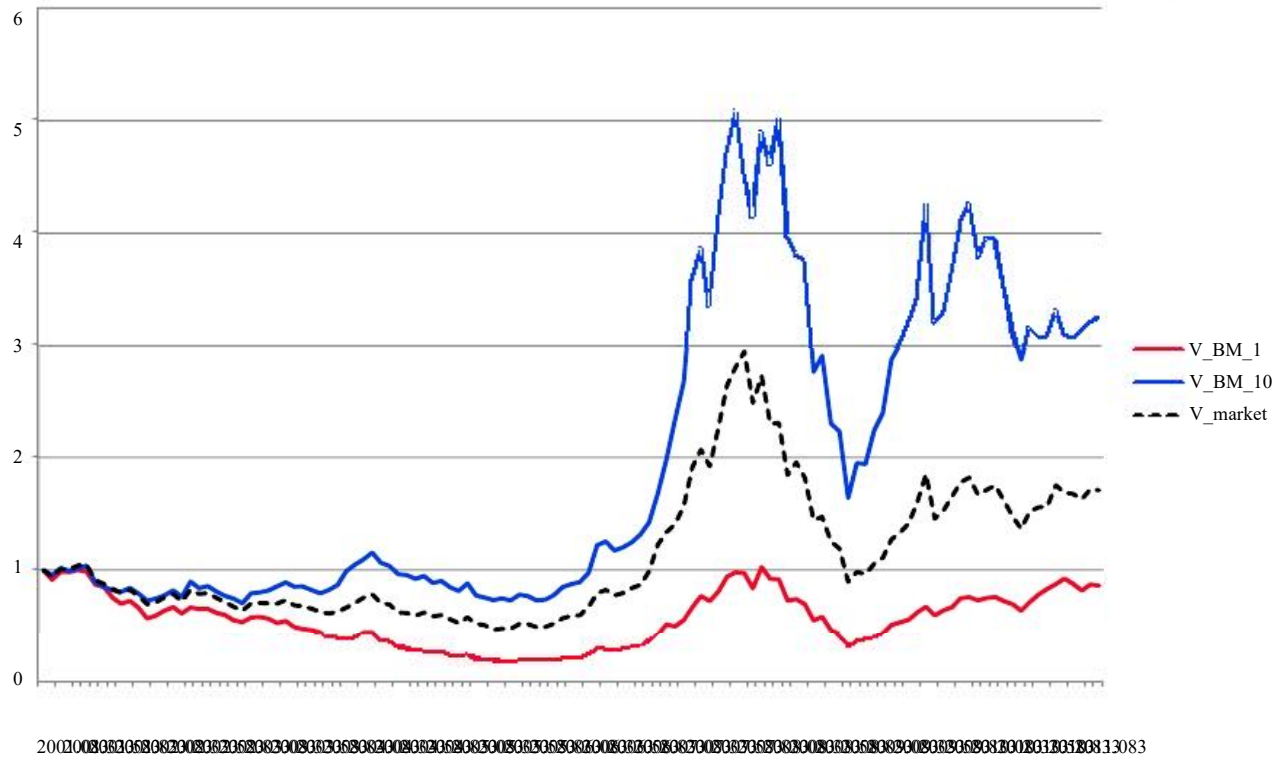
张劲帆

Value Stocks Won the Market: China (1995-2000)



张劲帆

Value Stocks Won the Market: China (2001-2010)



张劲帆

Stock Market Fluctuation and Market Expectation

☆ In the long run, stock must reflect a firm's profitability (the fundamentals of the firm)

☆ If the stock price deviates from its fundamental, it must return

– the larger the price fluctuation, the easier to estimate

– the stock price likes a randomly flying pollen, but in fact subject to gravity and will fall eventually

Long Term Investment and Return

- The relationship between stock market and economy is described by, Andre Kostolany, the German stock market Guru, as “master and his busy dog”. Every master with a dog has similar experiences, in which the dogs often run ahead of you during walking time, and then run back, and sometimes they are left behind. Finally, after arriving at home, the dogs may have run 4 km while the master may only walked 1km. There is a pattern that always holds, the dogs always run around their master.

Final Words on PE

E can be different: Static PE, Forward/dynamic PE, and Trailing PE

Pay attention to the quality of E: One time items must be adjusted: depreciation; research cost capitalization; cyclicality of earning